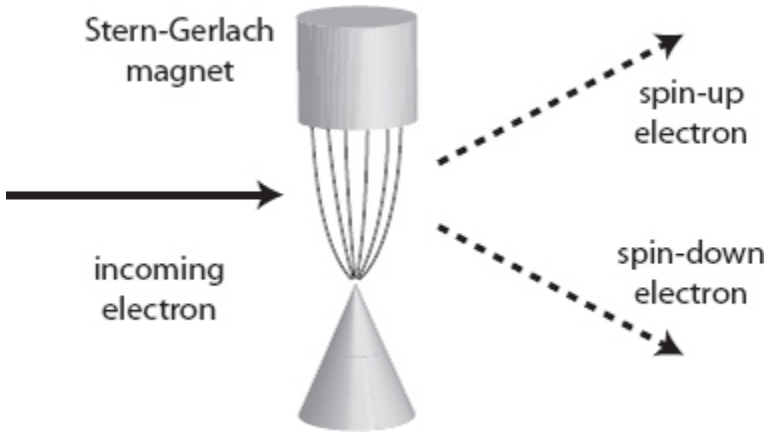


That seems surprising. Even if we've acclimated ourselves to the idea that the energy of an electron orbiting a nucleus only comes in certain quantized values, at least that energy seems like an objective property of the electron. But this thing we call the "spin" of the electron seems to give us different answers depending on how we measure it. No matter what particular direction we measure the spin along, there are only two possible outcomes we can obtain.

To make sure we haven't lost our minds, let's be clever and run the electron through two magnets in a row. Remember that the rules of textbook quantum mechanics tell us that if we get a certain measurement outcome, then measure the same system immediately again, we will always get the same answer. And indeed that's what happens; if an electron is deflected upward by one magnet (and is therefore spin-up), it will always be deflected upward by a following magnet oriented in the same way.

What if we rotate one of the magnets by 90 degrees? So we're splitting an initial beam of electrons into spin-up and spin-down as measured by a vertically oriented magnet, then taking the spin-up electrons and passing them through a horizontally oriented magnet. What happens then? Do they hold their breath and refuse to pass through, because they are vertically oriented spin-up electrons and we're forcing them to be measured along a horizontal axis?